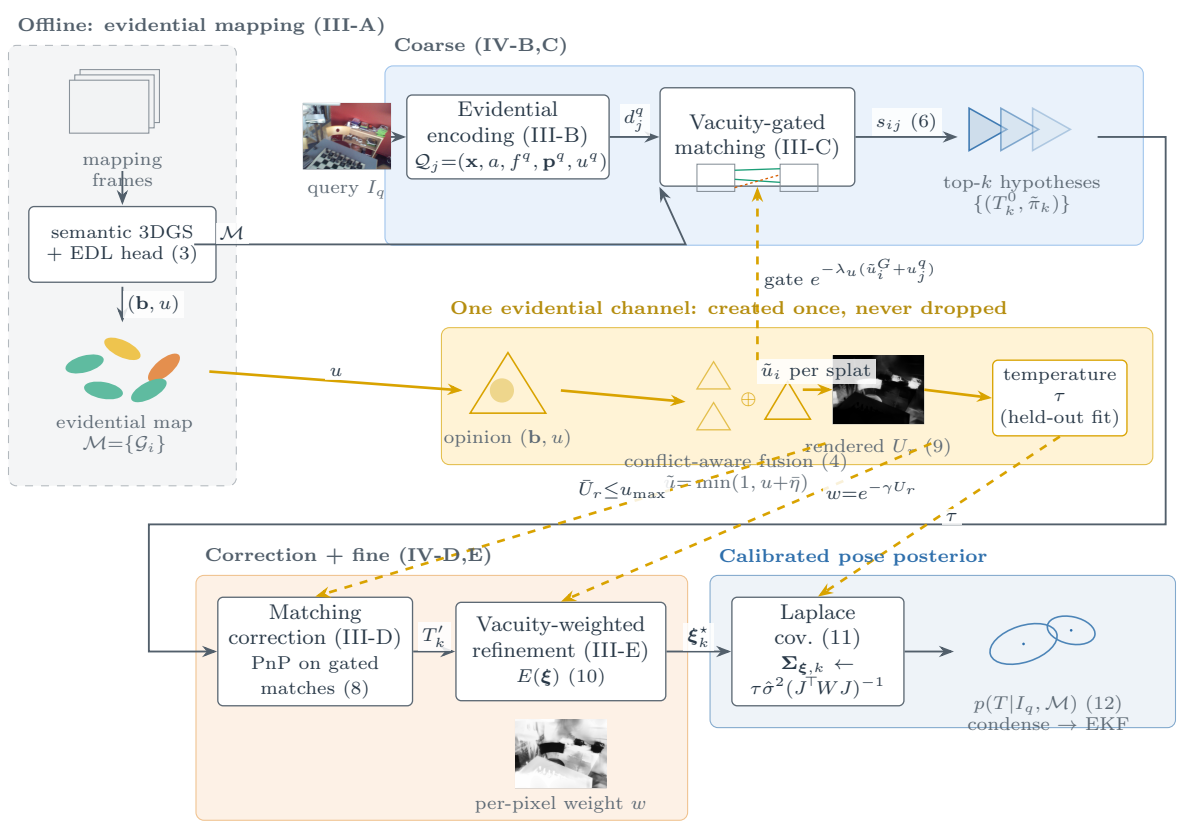

Offline: evidential mapping (III-A)
mapping frames
semantic 3DGS + EDL head (3)
$\mathcal{M}$
$(\mathbf{b}, u)$
evidential map $\mathcal{M}=\{\mathcal{G}_i\}$
Coarse (IV-B,C)
query $I_q$
Evidential encoding (III-B) $\mathcal{Q}_j=(\mathbf{x}, a, f^q, \mathbf{p}^q, u^q)$
$d_j^q$
Vacuity-gated matching (III-C)
$s_{ij}$ (6)
top-$k$ hypotheses $\{(T_k^0, \tilde{\pi}_k)\}$
gate $e^{-\lambda_u(\bar{u}_i^G + u_j^q)}$
One evidential channel: created once, never dropped
$u$
opinion $(\mathbf{b}, u)$
$\tilde{u}_i$ per splat
rendered $U_r$ (9)
temperature $\tau$ (held-out fit)
conflict-aware fusion (4)
$\bar{U}_r \le u_{\max}$
$\bar{\eta} = \min(1, u+\bar{\eta})$
$w = e^{-\gamma U_r}$
$\hat{\tau}$
Correction + fine (IV-D,E)
Matching correction (III-D) PnP on gated matches (8)
$T_k'$
Vacuity-weighted refinement (III-E) $E(\boldsymbol{\xi})$ (10)
$\boldsymbol{\xi}_k^\star$
per-pixel weight $w$
Calibrated pose posterior
Laplace cov. (11) $\boldsymbol{\Sigma}_{\boldsymbol{\xi},k} \leftarrow \tau\hat{\sigma}^2(J^\top W J)^{-1}$
$p(T|I_q, \mathcal{M})$ (12) condense → EKF